

ME302: 2025–26-II

Course Project Report

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1 Problem Description

A turbomachine component, designed for a future gas turbine engine by company X, is to be tested at a reduced geometric scale in a laboratory. The research laboratory Y operates a *closed-loop* high-pressure testing facility whose schematic is shown in Figure 1.

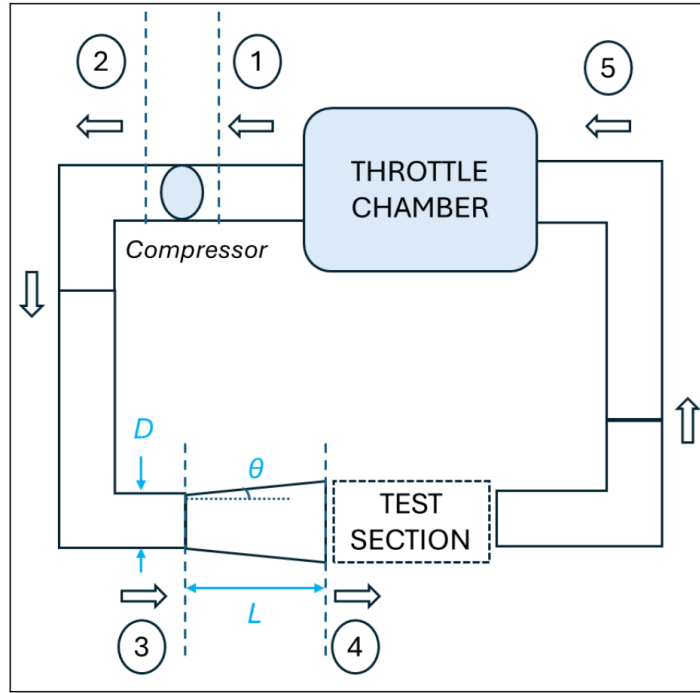


Figure 1: Closed-loop testing facility and station layout used in the analysis

Design objective. For the model to be representative of real engine conditions, both the **Mach number** and **Reynolds number** must be matched simultaneously at the test section:

$$M_{\text{target}} = 0.5, \quad Re_{\text{target}} = 3.0 \times 10^6 \quad (\text{based on } l_{\text{prototype}} = 0.2 \text{ m}).$$

The geometric scale is defined as:

$$s = \frac{l_{\text{model}}}{l_{\text{prototype}}}, \quad 0 < s \leq 1. \quad (1)$$

A larger scale makes instrumentation much easier to handle practically, so we want to maximize s . We seek the **maximum scale** s that satisfies all facility constraints.

Table 1: Given parameters for the analysis.

Parameter	Symbol	Value
Pipe diameter	D	0.6 m
Diffuser length	L	1.5 m
Prototype length	l_{proto}	0.2 m
Inlet stagnation temp.	T_{01}	293 K
Specific gas constant	R	287 J/(kg K)
Ratio of specific heats	γ	1.4
Dynamic viscosity	μ	1.83×10^{-5} Pa s
Reference pressure (ISA)	p_{ref}	101.325 kPa
Reference temperature	T_{ref}	288.15 K

Given data.

Facility constraints.

1. **Structural:** Minimum allowable pressure anywhere in the facility, $p_{\text{min}} = 200$ kPa.
2. **Closed-loop operation:** $p_{05} \geq p_{01}$, otherwise the inlet conditions cannot be maintained and the closed-loop flow will not sustain.

2 Methodology

2.1 Governing Equations

2.1.1 Isentropic Flow Relations for an Ideal Gas

For a calorically perfect ideal gas with ratio of specific heats γ , the isentropic relations between static and stagnation (total) quantities at a given Mach number M are:

$$\frac{T}{T_0} = \frac{1}{1 + \frac{\gamma - 1}{2} M^2} \quad (2)$$

$$\frac{p}{p_0} = \left(\frac{T}{T_0} \right)^{\gamma/(\gamma-1)} = \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{-\gamma/(\gamma-1)} \quad (3)$$

$$\frac{\rho}{\rho_0} = \left(\frac{T}{T_0} \right)^{1/(\gamma-1)} = \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{-1/(\gamma-1)} \quad (4)$$

The speed of sound in the gas is:

$$a = \sqrt{\gamma R T} \quad (5)$$

2.1.2 Reynolds Number

The Reynolds number at the test section (station 4) based on the model length l_{model} :

$$Re = \frac{\rho_4 V_4 l_{\text{model}}}{\mu} \quad (6)$$

where ρ_4 is the static density, $V_4 = M_4 \cdot a_4$ is the flow velocity, and μ is the dynamic viscosity.

2.1.3 Corrected (Reference) Mass Flow Rate

The compressor map provides the *corrected* mass flow rate $\dot{m}_{\text{ref}}$ at reference conditions $(p_{\text{ref}}, T_{\text{ref}})$. The actual mass flow rate is:

$$\dot{m} = \dot{m}_{\text{ref}} \frac{p_{01}/p_{\text{ref}}}{\sqrt{T_{01}/T_{\text{ref}}}} \quad (7)$$

2.1.4 Continuity Equation at the Test Section

The mass flow rate from continuity at station 4:

$$\dot{m} = \rho_4 V_4 A_4 \quad (8)$$

where $A_4 = \frac{\pi}{4} D_4^2$ is the test section cross-sectional area.

2.1.5 Diffuser / Nozzle Loss Coefficient

The total pressure loss coefficient for the conical duct between stations 3 and 4 is defined as:

$$K = \frac{p_{03} - p_{04}}{\frac{1}{2} \rho_3 V_3^2} \quad (9)$$

with:

$$K = \begin{cases} 0 & \text{if semi-angle } \theta \leq 0 \text{ (converging nozzle)} \\ 3.2 \tan^{3/2} \theta \left(1 - \left(\frac{A_3}{A_4} \right)^2 \right) & \text{if } \theta > 0 \text{ (diverging diffuser)} \end{cases} \quad (10)$$

where the semi-angle is:

$$\theta = \arctan \left(\frac{D_4/2 - D/2}{L} \right) \quad (11)$$

2.2 Station-by-Station Analysis

2.2.1 Geometric Relations

Given a scale s :

$$l_{\text{model}} = s \times l_{\text{prototype}} = 0.2 s \text{ [m]} \quad (12)$$

$$D_4 = 2 l_{\text{model}} = 0.4 s \text{ [m]} \quad (\text{sufficient gap for instrumentation}) \quad (13)$$

$$A_4 = \frac{\pi}{4} D_4^2 = \frac{\pi}{4} (0.4 s)^2 = 0.04 \pi s^2 \text{ [m}^2\text{]} \quad (14)$$

2.2.2 Diffuser Semi-Angle Check

Since $D_4 = 0.4 s \leq 0.4 < 0.6 = D$ for all $s \in (0, 1]$:

$$\theta = \arctan\left(\frac{0.2s - 0.3}{1.5}\right) < 0 \quad \forall s \in (0, 1] \quad (15)$$

The duct between stations 3 and 4 is always a **converging nozzle**, so $K = 0$ and there is *no stagnation pressure loss*:

$$p_{04} = p_{03} \quad (16)$$

2.2.3 Stagnation Temperature Distribution

The compressor raises the stagnation temperature:

$$T_{02} = (T_0 \text{ ratio}) \times T_{01} \quad (17)$$

Since the facility is **adiabatic** beyond station 2:

$$T_{02} = T_{03} = T_{04} = T_{05} \quad (18)$$

2.2.4 Stagnation Pressure Distribution

The pressure chain through the facility:

$$p_{02} = (p_0 \text{ ratio}) \times p_{01} \quad (19)$$

$$p_{03} = p_{02} - 0.015 p_{02} = 0.985 p_{02} \quad (20)$$

$$p_{04} = p_{03} \quad (K = 0, \text{ see Eq. (16)}) \quad (21)$$

$$p_{05} = p_{04} - 0.05 p_{04} = 0.95 p_{04} \quad (22)$$

Combining:

$$p_{05} = 0.95 \times 0.985 \times (p_0 \text{ ratio}) \times p_{01} = 0.93575 \times (p_0 \text{ ratio}) \times p_{01} \quad (23)$$

2.2.5 Test Section Conditions (Station 4)

With $M_4 = 0.5$ and $\gamma = 1.4$:

$$\frac{T_4}{T_{04}} = \frac{1}{1 + 0.2 \times 0.25} = \frac{1}{1.05} = 0.95238 \quad (24)$$

$$\frac{p_4}{p_{04}} = (1.05)^{-3.5} = 0.84302 \quad (25)$$

$$\frac{\rho_4}{\rho_{04}} = (1.05)^{-2.5} = 0.88517 \quad (26)$$

The static temperature, speed of sound, and velocity:

$$T_4 = \frac{T_{04}}{1.05} \quad (27)$$

$$a_4 = \sqrt{\gamma R T_4} = \sqrt{1.4 \times 287 \times T_4} \quad (28)$$

$$V_4 = M_4 \times a_4 = 0.5 a_4 \quad (29)$$

2.2.6 Density and Pressure from Reynolds Number

Rearranging Eq. (6):

$$\rho_4 = \frac{Re \cdot \mu}{V_4 \cdot l_{\text{model}}} = \frac{3 \times 10^6 \times 1.83 \times 10^{-5}}{V_4 \times 0.2 \text{ s}} = \frac{54.9}{V_4 \times 0.2 \text{ s}} = \frac{274.5}{V_4 \cdot s} \quad (30)$$

From the ideal gas law:

$$p_4 = \rho_4 R T_4 \quad (31)$$

The stagnation pressure at station 4:

$$p_{04} = \frac{p_4}{0.84302} \quad (32)$$

And the inlet stagnation pressure:

$$p_{01} = \frac{p_{04}}{0.985 \times (p_0 \text{ ratio})} \quad (33)$$

2.2.7 Mass Flow Rate

From continuity (Eq. (8)):

$$\dot{m}_{\text{continuity}} = \rho_4 V_4 A_4 \quad (34)$$

Substituting ρ_4 from Eq. (30):

$$\dot{m}_{\text{continuity}} = \frac{274.5}{V_4 \cdot s} \times V_4 \times 0.04\pi s^2 = 274.5 \times 0.04\pi \times s = \underbrace{0.2\pi \times Re \times \mu \times s}_{=34.51} \quad (35)$$

Note that $\dot{m}_{\text{continuity}}$ depends *linearly* on scale and is **independent of the compressor operating point** (the velocity terms cancel exactly).

From the compressor (Eq. (7)):

$$\dot{m}_{\text{compressor}} = \dot{m}_{\text{ref}} \times \frac{p_{01}/p_{\text{ref}}}{\sqrt{T_{01}/T_{\text{ref}}}} \quad (36)$$

2.2.8 Mass Flow Balance Equation

Setting $\dot{m}_{\text{continuity}} = \dot{m}_{\text{compressor}}$:

$$\boxed{34.51 \times s = \dot{m}_{\text{ref}} \times \frac{p_{01}(s)}{p_{\text{ref}} \cdot \sqrt{T_{01}/T_{\text{ref}}}}} \quad (37)$$

where $p_{01}(s)$ itself depends on s through Eqs. (30)–(33). This is a **nonlinear equation in s** for each compressor operating point.

2.3 Solution Algorithm

We've laid out the full step-by-step procedure we followed in Algorithm 1.

Algorithm 1 Finding the maximum feasible scale

Require: Compressor data: $\{(\dot{m}_{\text{ref},i}, T_{0,\text{ratio},i}, p_{0,\text{ratio},i})\}_{i=1}^N$

Ensure: Maximum scale s^* and corresponding operating point

```
1:  $s^* \leftarrow 0$ 
2: for  $i = 1$  to  $N$  do ▷ Loop over all compressor points
3:   Define residual function  $f(s)$  for operating point  $i$ :
      $T_{04} = T_{0,\text{ratio},i} \times T_{01}$ 
      $T_4 = T_{04}/1.05$ 
      $V_4 = 0.5 \times \sqrt{\gamma R T_4}$ 
      $\rho_4 = Re \cdot \mu / (V_4 \times 0.2 \text{ s})$ 
      $p_4 = \rho_4 R T_4$ 
      $p_{04} = p_4 / 0.84302$ 
      $p_{01} = p_{04} / (0.985 \times p_{0,\text{ratio},i})$ 
      $\dot{m}_{\text{cont}} = \rho_4 V_4 A_4(s)$ 
      $\dot{m}_{\text{comp}} = \dot{m}_{\text{ref},i} \times p_{01} / (p_{\text{ref}} \sqrt{T_{01}/T_{\text{ref}}})$ 
      $f(s) = \dot{m}_{\text{cont}} - \dot{m}_{\text{comp}}$ 
4:   Solve  $f(s) = 0$  for  $s \in (0.01, 1.0)$  using Brent's method
5:   Evaluate constraints at solution  $s_i$ :
      $p_{05} = 0.95 \times p_{04} \geq p_{01}$ ?
      $p_4 \geq 200 \text{ kPa}$ ?
      $p_{01} \geq 200 \text{ kPa}$ ?
6:   if all constraints satisfied and  $s_i > s^*$  then
7:      $s^* \leftarrow s_i$ ; record operating point  $i$ 
8:   end if
9: end for
10: return  $s^*$  and corresponding operating point
```

Implementation. The algorithm is implemented in Python using NumPy for numerical computations and `scipy.optimize.brentq` for robust root-finding. The Brent method is chosen because the residual function changes sign between $s = 0.01$ and $s = 1.0$, guaranteeing convergence. A tolerance of 10^{-10} is used to ensure high precision. The code is organised as a Jupyter notebook (`ME302_project.ipynb`).

3 Results and Discussion

3.1 Scale Distribution Across Operating Points

When we ran through all 138 operating points every single one gave a valid solution, with scales falling in a smaller range between roughly 0.60 and 0.64. Also, *all* 138 points satisfy the structural minimum-pressure constraint and the closedloop constraint.

3.2 Maximum Feasible Scale

The **maximum feasible scale** is:

$$\boxed{s_{\text{max}} = 0.6358} \tag{38}$$

This corresponds to a model length of $l_{\text{model}} = 127.16$ mm and a test-section diameter of $D_4 = 254.32$ mm.

3.3 Optimal Compressor Operating Point

The maximum scale is achieved at the first data point in the compressor map (highest corrected flow, lowest pressure ratio):

Table 2: Optimal compressor operating point.

Quantity	Value
$\dot{m}_{\text{ref}}$	10.5503 kg/s
T_0 ratio (T_{02}/T_{01})	1.0685
p_0 ratio (p_{02}/p_{01})	1.2105

3.4 Complete Station Conditions

Table 3 lists all stagnation and static quantities at the optimal operating point.

Table 3: Station-by-station conditions at $s = 0.6358$.

Station	T_0 [K]	p_0 [kPa]	Notes
1 (Compressor inlet)	293.00	212.39	Given T_{01}
2 (Compressor outlet)	313.08	257.09	$\times T_0$ ratio, $\times p_0$ ratio
3 (Diffuser inlet)	313.08	253.24	-1.5% piping loss
4 (Test section)	313.08	253.24	$K = 0$ (nozzle)
5 (Downstream exit)	313.08	240.58	-5% downstream loss
<i>Static conditions at test section (station 4):</i>			
	$T_4 = 298.17$ K	$p_4 = 213.48$ kPa	$\rho_4 = 2.495$ kg/m ³
	$V_4 = 173.06$ m/s	$\dot{m} = 21.93$ kg/s	$M_4 = 0.5$

3.5 Sample Calculations

For verification, the key intermediate results at $s = 0.6358$ are computed step by step:

1. Geometry:

$$l_{\text{model}} = 0.6358 \times 0.2 = 0.12716 \text{ m}, \quad D_4 = 2 \times 0.12716 = 0.25432 \text{ m}$$

$$A_4 = \frac{\pi}{4}(0.25432)^2 = 0.05080 \text{ m}^2$$

2. Stagnation temperature:

$$T_{04} = 1.0685 \times 293 = 313.08 \text{ K}$$

3. Static conditions at $M_4 = 0.5$:

$$T_4 = \frac{313.08}{1.05} = 298.17 \text{ K}, \quad a_4 = \sqrt{1.4 \times 287 \times 298.17} = 346.13 \text{ m/s}$$

$$V_4 = 0.5 \times 346.13 = 173.06 \text{ m/s}$$

4. Density from Reynolds number:

$$\rho_4 = \frac{3 \times 10^6 \times 1.83 \times 10^{-5}}{173.06 \times 0.12716} = \frac{54.9}{22.01} = 2.495 \text{ kg/m}^3$$

5. Pressures:

$$p_4 = 2.495 \times 287 \times 298.17 = 213\,484 \text{ Pa} = 213.48 \text{ kPa}$$

$$p_{04} = \frac{213.48}{0.84302} = 253.24 \text{ kPa}$$

$$p_{02} = \frac{253.24}{0.985} = 257.09 \text{ kPa}, \quad p_{01} = \frac{257.09}{1.2105} = 212.39 \text{ kPa}$$

$$p_{05} = 0.95 \times 253.24 = 240.58 \text{ kPa}$$

6. Mass flow verification:

$$\dot{m}_{\text{cont}} = 2.495 \times 173.06 \times 0.05080 = 21.93 \text{ kg/s}$$

$$\dot{m}_{\text{comp}} = 10.5503 \times \frac{212394/101325}{\sqrt{293/288.15}} = 10.5503 \times \frac{2.0961}{1.00838} = 21.93 \text{ kg/s} \checkmark$$

3.6 Constraint Verification

Table 4: Constraint verification at the optimal operating point.

Constraint	Required	Computed	Status
Reynolds number Re	3.0×10^6	3,000,000.0	✓
Mach number M_4	0.5	0.5000	✓
Min. pressure p_4	$\geq 200 \text{ kPa}$	213.48 kPa	✓
Min. pressure p_{01}	$\geq 200 \text{ kPa}$	212.39 kPa	✓
Closed loop $p_{05} \geq p_{01}$	—	$240.58 \geq 212.39$	✓
Mass flow balance	0	$1.9 \times 10^{-10} \text{ kg/s}$	✓

3.7 Compressor Operating Map

Figure 2 shows the compressor operating map with the selected operating point highlighted. The optimal point corresponds to the highest corrected mass flow rate and the lowest pressure ratio on the map.

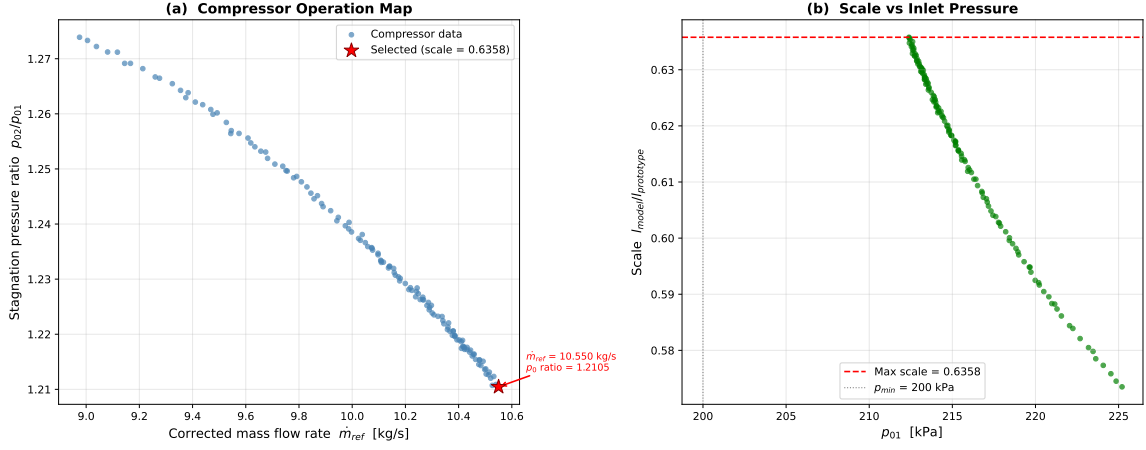


Figure 2: (a) Compressor operating map (p_0 ratio vs $\dot{m}_{ref}$) with the selected operating point marked by a red star. (b) Computed scale for each operating point as a function of inlet stagnation pressure p_{01} ; green points satisfy all constraints.

3.8 Discussion

Scale-limiting mechanism. What ended up capping the scale was the 200 kPa pressure floor not the closed-loop condition as we initially expected. To understand this, consider how scale affects pressures:

- As scale s increases, $l_{model} = 0.2 s$ increases.
- From the Reynolds number constraint (Eq. (30)), $\rho_4 \propto 1/s$, so density *decreases*.
- Implies, $p_4 \propto 1/s$ and $p_{01} \propto 1/s$ also decrease.
- Finally, p_{01} or p_4 approaches the 200 kPa structural limit.

At the optimal point, $p_{01} = 212.39$ kPa and $p_4 = 213.48$ kPa—both only $\sim 6\%$ above the limit, which tells us we’re cutting it pretty close to the 200 kPa limit.

Why the highest $\dot{m}_{ref}$ point is optimal. At higher flow, lower pressure-ratio points, the compressor doesn’t need to work as hard, so pressures across the system stay higher and we get more headroom before hitting the 200 kPa floor. That’s essentially why the first data point on the compressor map turned out to be the best one for our purposes.

Closedloop constraint margin. The closed-loop constraint $p_{05} \geq p_{01}$ is satisfied with the fact that there is a gap of about 28.18 kPa here, so this constraint wasn’t what actually limited us.

4 Supplement

One more file is submitted with this report:

1. ME302.ipynb - Jupyter notebook (Python 3) containing the complete analysis code with:
 - Data loading and preprocessing
 - Isentropic flow functions

- Station-by-station computation
- Brent root-finding for mass-flow balance
- Constraint checking
- Result tables and verification
- Plot generation

Software dependencies. The code requires Python 3 with `numpy`, `scipy`, and `matplotlib`. All computations were verified to produce results matching the hand calculations presented in Section 3.5.